

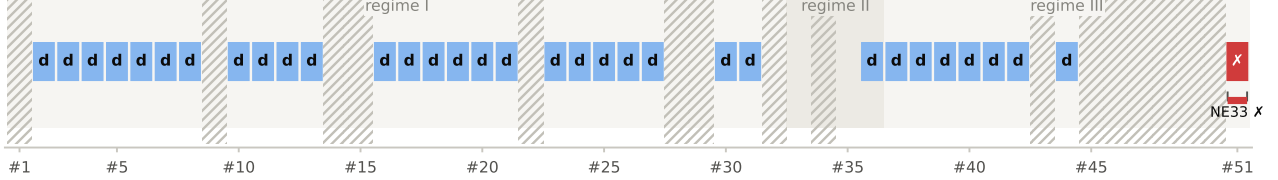
H81: Culture beyond composition: the emergent slow mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. In regime I the village content has a collective slow mode ($\tau_u \approx 25$ days) beyond its members and the measured goal and operator fields, carried across turnover; it does not follow the long-lived artifact record its members read, and an unmeasured calendar-time drift remains the open rival.

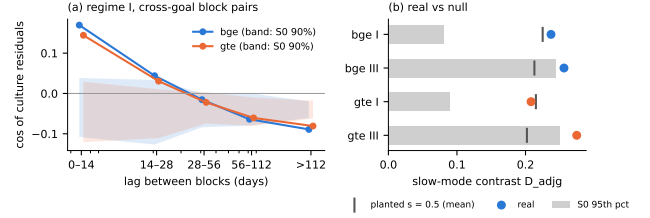
level: descriptive · pipeline checks 86% · updated 2026-10-04



Goal periods tested (2 failed, 32 descriptive). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The test asks whether village content carries a slow component that its members do not explain. The null is each member's personal vector (two-way fixed effects on its other goal periods) plus the measured goal fields. In regime I (24 goals) culture residuals of blocks ≤ 14 days apart align ($\bar{s} = 0.17$), decaying with $\tau_u \approx 23$ –28 days; the contrast is 3 times the synthetic null in both embedding models and survives turnover. Round 2 asked what carries it. It does not follow the long-lived artifact record its members read ($C_{R1} = +0.02 / -0.07$ against an exogenous-drift 95th percentile of 0.11/0.09, power 1.0). It survives removing operator directions beyond the centroid and is not concentrated in outside-world talk. H82's boundary trace fits the same τ . Calendar versus village time cannot be identified, so outside drift is narrowed, not beaten.



Observables. (a) Regime I: mean similarity of culture residuals between blocks of different goal periods against their lag. Bands are the 90% range of the synthetic composition null (S0) on the real panel. Similarity is high below two weeks and decays over about a month. (b) The slow-mode contrast D_{adj} per model and regime (dots), the S0 95th percentile (grey bars) and the planted mean for a slow mode with half the period-field variance (ticks).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect a village to drift as a whole on a timescale of about a month, independent of who is in it. In regime I the slow-mode contrast matches a synthetic mode that carries half of the period-field variance ($s = 0.5$). That share is inferred from the match, not measured. A new kickoff does not reset the village: neighbouring goals share about 0.17 of their residual direction. Roster changes do not move the mode. Its carrier is still unknown, but it is not the long-lived artifact record (old sites, documents, repos), the operator's messages or outside-world talk, so curating those artifacts will not steer it. For monitoring, compare a village with its own last two to three weeks, not with its whole history.

PHYSICAL MODEL

Vector spins (model 11). The spin $v_{i,d}$ is agent i 's style-residualized content vector on day d (32-d). a_i is the personal vector (weights, family, style). h_G is the exogenous field of goal period G , removed by projection Π_G . w_b is a topic field with no memory. u_b is the collective slow mode, an Ornstein–Uhlenbeck vector with correlation time τ_u . The order parameter is the block culture residual u_b . The statistic D is the near-minus-far similarity of u_b across goal boundaries, adjusted for goal similarity (D_{adj}) and member overlap (D_{adj}).

$$v_{i,d} = a_i + h_{G(d)} + w_{b(d)} + u_{b(d)} + e_{i,d}, \quad \langle u_b \cdot u_{b'} \rangle \propto e^{-|t_b - t_{b'}|/\tau_u}$$

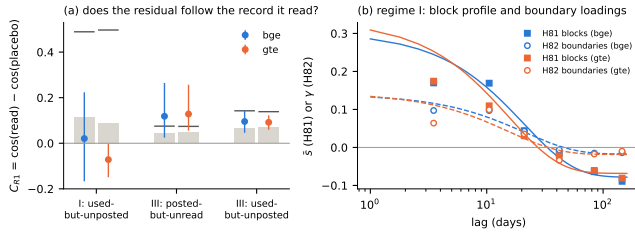
$$r_{i,b} = \Pi_G(\bar{v}_{i,b} - \hat{a}_i^{(-G)}), \quad u_b = \langle r_{i,b} \rangle_{i \in b}$$

$$D = \langle \cos(u_b, u_{b'}) \rangle_{\Delta t \leq 14 \text{ d}} - \langle \cos(u_b, u_{b'}) \rangle_{\Delta t \geq 42 \text{ d}}$$

H81-culture-beyond-composition · generated 2026-10-04 14:49 by in-fra/summaries/build_summaries.py · card hypotheses/H81-culture-beyond-composition/README.md · ratings: H81 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction	Observed (bge / gte)	Verdict
P2 slow mode $D_{\text{adjg}} > S_0$ q95	I: 0.237 / 0.208 (q95 0.081 / 0.089); $\tau_u = 28 \pm 5$ / 23 ± 4 d	pass (I)
P3 across turnover (D_{adj})	I: 0.166 / 0.170 (q95 0.10)	pass (I)
P4 roster jumps; N1 #51; N2 NE33	percentile 0.08 / 0.14; $L_{\text{slow}} < 0$; 0.14–0.89	fail
P5 without Gemini 2.5 Pro	$\times 1.06$; size-matched $\times 0.90$ / $\times 0.96$	pass
R1 follows read record (I)	$C_{R1} = +0.02$ / -0.07 (q95 0.11 / 0.09; power 1.0)	fail
R1 regime III (descr.)	$+0.12$ / $+0.13$ (q95 0.04 / 0.05)	pass
R2a operator beyond centroid	D_{adjg} 0.244 / 0.229 after removal (q95 0.07 / 0.08)	survives
R2b outside-world topics	$L_{\text{out}} = +0.004$ / $+0.009$ (q95 0.018 / 0.014)	none
R2c calendar vs village clock	goal count lowest SSE ($\tau = 2.4$ / 1.7 goals); not identifiable	inconcl.
R4 one OU with H82	$\Delta\tau = -12$ / -11 d (band $[-33, +6]$); joint $\tau = 21$ / 16 d (weak)	pass



Second figure. Round 2. (a) Record-carrier contrast C_{R1} (read minus placebo record content) with goal-bootstrap 95% CIs; grey bars are the exogenous-drift (S_{ex}) 95th percentile, ticks the planted record-carriage mean ($\lambda = 1$). Regime I fails; regime III passes. (b) Regime I: H81 block similarity (squares) and H82 boundary loadings (circles) with the shared- τ OU fit.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive

+24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only measured exogenous directions are removed; a calendar-time drive reaching every agent without operator messages or outside-world talk would look the same, and the clock test cannot tell calendar from village time (synthetic accuracy ≤ 0.64). The regime-I artifact channel is thin (188 old-artifact chat mentions in 9 months). Regime-III record-following may partly be room topics (3–5 goals). The H82 one-mode band is wide. Power is 0.96–0.99 for a slow mode with half the period-field variance. τ_u is a descriptive fit.

ROUND 2

Test the rolling chat window (each block reads the previous block's tail) as the carrier. Use regime-I newcomers to separate calendar drift (immediate) from a village mode (needs reading). Redo regime-III record-following within rooms at matched lag.

- **Essence.** a village-level state variable that persists for weeks and survives turnover, and what carries it.
- **H81-R1** Name the carrier.
- **H81-R2** Exogenous drift probe.
- **H81-R3** Regime III with more units.
- **H81-R4** Link to H82.